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Applying Probability and Statistics to Evaluate a Toxic Comment Classifier

Classifying toxic comments with BERT and Twitter-RoBERTa, and using statistics to test whether the difference between them is real

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Summary

Our goal in this project was to build a classifier that automatically detects toxic comments and tweets, and to understand how different language models handle this task. As a team, we split the work: Abhishek fine-tuned and analysed Twitter-RoBERTa, while Rahul and Shashwath worked on BERT. Both models were trained on the same data with the same settings, and at the end we brought our results together and compared them on the same 20,000 test comments.

Both models learned the task well: each classifies comments with about 95% accuracy and a ROC-AUC above 0.96. Twitter-RoBERTa scored slightly higher on F1 (0.6923 against BERT's 0.6840). But when we put a 95% confidence interval around that 0.83-point gap, it runs from −0.43 to +2.10 points, and a permutation test gives p = 0.19. So on F1, we cannot rule out that the gap is just the luck of which comments ended up in the test set.

On ROC-AUC (p = 0.004) and on the overall error rate (McNemar's test, p = 0.015), RoBERTa's small edge is statistically significant. Our conclusion is that both models are effective toxic-comment classifiers and RoBERTa is probably a little better, but the improvement is too small to show up reliably in F1 with a test set of this size.

1. Problem Statement and Dataset

Why this problem matters

Anyone who spends time on social media has seen how quickly a thread can turn nasty. Platforms try to catch toxic posts automatically, because human moderators cannot read millions of comments a day, and every hour a toxic post stays up, it reaches more people. Choosing a good classifier therefore matters.

Our main aim was practical: build a model that can classify a comment or tweet as toxic or not, and understand how modern language models make that decision. Along the way, a second question came up. When one model scores 1% higher than another, that number comes from one specific test set, and a different test set might have produced a different winner. So, once both models were trained, we asked:

> *Does the improved model actually perform better, or could the difference we observed be due to variation in the test sample?*

The two models

We worked on two models in parallel, chosen to be as similar as possible except for one thing.

Role	Model	Size	What it learned from before fine-tuning	Worked on by
Baseline	BERT-base (bert-base-uncased)	110 million parameters	Books and English Wikipedia	Rahul Kemburu, Shashwath Naren PV

Role	Model	Size	What it learned from before fine-tuning	Worked on by
Improved	Twitter-RoBERTa (cardiffnlp/twitter-roberta-base-2022-154m)	125 million parameters	General web text, then 154 million tweets	Abhishek Dadi

Both are transformer models of almost the same size. The real difference is what they read during pre-training. BERT learned English from books and Wikipedia, while Twitter-RoBERTa was trained further on 154 million tweets. Toxic comments are full of slang, abbreviations and creative spelling, so going in, we expected the tweet-trained model to have an edge.

We fine-tuned both models in exactly the same way, on the same data, with the same settings. That way, if one of them comes out ahead, the difference in pre-training is the reason, not the training setup.

The dataset

We used Civil Comments, a public dataset released by Jigsaw and Google under a CC0 licence and loaded from Hugging Face (google/civil_comments). It contains about two million real comments posted on news websites, each rated by several human annotators. The official splits have 1,804,874 training, 97,320 validation and 97,320 test comments.

Every comment comes with a toxicity score: the fraction of annotators who considered it toxic. We labelled a comment as toxic if at least half of the annotators said so (score ≥ 0.5). This turns the task into a simple yes-or-no classification.

One thing stood out immediately: toxic comments are rare. Only 7.99% of the test comments are toxic. That has a big effect on evaluation. A lazy model that always answers "not toxic" would already be 92% accurate while being completely useless. For that reason, we chose the F1 score for the toxic class as our main metric and kept accuracy and ROC-AUC as supporting measures.

2. Sampling and Experimental Methodology

The idea behind the experiment is simple: treat both models fairly, test them on exactly the same comments, and then ask how much the result could change if those test comments had been different. Every random step used the same seed (42), and we saved the sampled data and every prediction, so anyone can rerun our code and get the same numbers.

2.1 How we sampled the data

Training on all 1.8 million comments would have taken many hours on the free Colab GPU, so we drew samples from each official split. Each sample was drawn differently, for a different reason.

Sample	Drawn from	Sampling method	Size	Toxic share
Training	Official train split	Disproportionate stratified	100,000	25% (chosen

Sample	Drawn from	Sampling method	Size	Toxic share
	(1,804,874)	sampling		on purpose)
Validation	Official validation split (97,320)	Simple random sampling	15,000	7.82%
Test	Official test split (97,320)	Proportionate stratified random sampling	20,000	7.99%

For training, we deliberately over-sampled toxic comments to 25%, so that the models would see plenty of examples of what they are supposed to detect. For validation and testing, we did the opposite and kept the natural rate of about 8%, because these sets need to look like the real world the model will face.

The test sample was stratified: we sampled toxic and non-toxic comments separately, in their true proportions, so the test set mirrors the population. Because each sample came from a different official split, no comment appears in more than one of them.

2.2 Did the sampling work?

Before trusting the samples, we checked whether their toxic share matched the full split they came from, using a one-sample z-test for a proportion. The null hypothesis is that the sample proportion equals the population proportion.

Sample	n	Toxic share in sample	Toxic share in population	z	p-value
Validation	15,000	0.0782	0.0788	−0.283	0.777
Test	20,000	0.0799	0.0799	−0.006	0.995

Both p-values are very high, so there is no evidence that either sample is unrepresentative. The test sample matches its population almost exactly, which is what stratified sampling is supposed to achieve.

2.3 Training both models

Both models got identical treatment: 2 epochs, batch size 32, learning rate 2×10^{-5} with a 6% warm-up, weight decay 0.01, and comments cut off at 128 tokens. Training ran on a single NVIDIA T4 GPU in Google Colab and took about 25 minutes per model. Before training, we did some light cleaning: web links became "http" and usernames became "@user".

Each model outputs a probability that a comment is toxic, and we still had to decide where to draw the line. Because the training data was 25% toxic, the models overestimate toxicity on real data, where only 8% is toxic, so the usual cut-off of 0.5 would flag far too much. Instead, we picked the threshold that maximised F1 on the validation set: 0.84 for BERT and 0.86 for RoBERTa.

The test set was used only once, for the final evaluation. It played no part in any of these choices.

2.4 Hypotheses

We fixed F1 as the main metric before looking at any results. Otherwise, it would be tempting to pick whichever metric happened to look best afterwards. Writing ΔF1 = F1(improved) − F1(baseline):

- H_0: ΔF1 = 0. The models are equally good, and any gap comes from the particular test sample.
- H_1: ΔF1 ≠ 0. One model is genuinely better.

We used a two-sided test at a significance level of α = 0.05. Accuracy and ROC-AUC were tested as supporting evidence.

2.5 Confidence intervals

A single number like "F1 = 0.69" hides how uncertain it is, so we gave every metric a 95% confidence interval.

Accuracy is the easiest case. Each test comment is either classified correctly or not, which is a Bernoulli trial. The number of correct answers therefore follows a binomial distribution, and we used the Wilson score interval, which stays accurate even when the proportion is close to 0 or 1.

F1 and ROC-AUC do not have a neat formula like this, so we used the bootstrap. The idea is to treat the test set as if it were the population, draw 20,000 comments from it with replacement, recompute the metric, and repeat this 10,000 times (2,000 times for ROC-AUC, which is slower to compute). The middle 95% of those values forms the confidence interval.

For the difference between the models, we used a paired bootstrap: in each resample, both models are scored on the same drawn comments. This matters because both models tend to find the same comments easy or hard, and pairing removes that shared noise.

2.6 Hypothesis tests

We used three different tests, so that the conclusion would not rest on a single method.

1. Permutation test on F1 (the main test). If H_0 is true, it should not matter which model's prediction we call "baseline" and which "improved". So for each comment, we randomly swapped the two models' predictions, recomputed ΔF1, and repeated this 10,000 times. This shows what ΔF1 looks like when there is truly no difference. The p-value is the share of shuffled results at least as extreme as the real one.
2. McNemar's test on error rates. Comments that both models get right, or both get wrong, say nothing about which model is better. Only the disagreements count: b is the number of comments only the baseline got right, and c the number only the improved model got right. If the models are equally good, b and c should be roughly equal. The test measures how far apart b and c are relative to their total, and compares this with a chi-square distribution with one degree of freedom.
3. A z-test using the bootstrap standard error. By the central limit theorem, ΔF1 is roughly normally distributed, so dividing the observed difference by its standard error gives a z-score and a quick, independent check.

2.7 Simulating other test sets

Finally, we tackled the research question head-on. We cannot collect thousands of new test sets, but we can simulate them. For eight test-set sizes, from 100 to 20,000 comments, we drew 1,000 random test sets each and scored both models on every one. We recorded how often RoBERTa came out on top, and how often McNemar's test would have called the difference significant, which is the test's power.

3. Observations, Decision, and Limitations

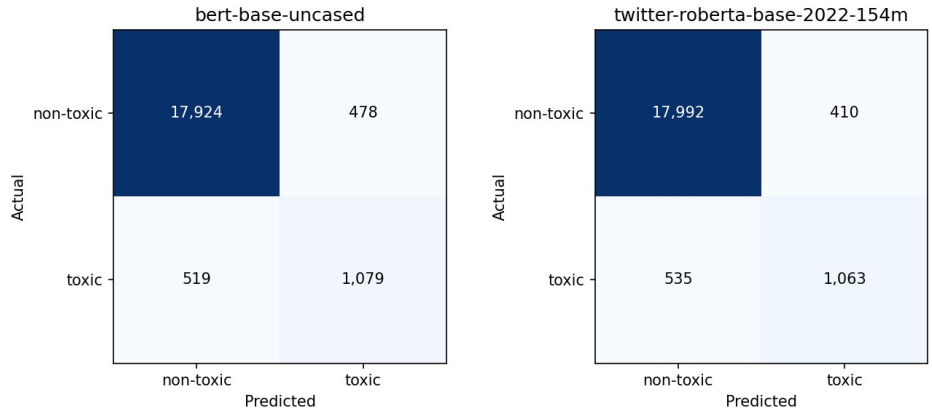
The short version: RoBERTa does a little better on almost everything, but on F1, our main metric, its advantage is too small to tell apart from random variation in the test set.

3.1 How each model did

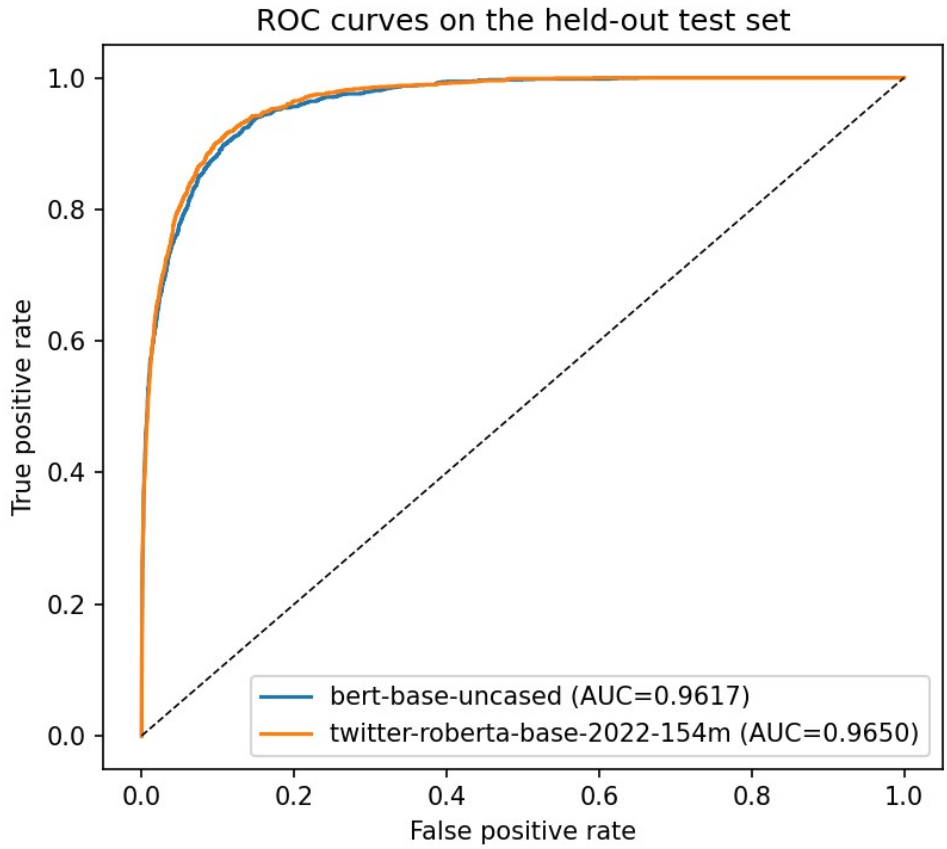
The test set has 20,000 comments, of which 1,598 (7.99%) are toxic. The numbers in brackets are 95% bootstrap confidence intervals.

Metric	BERT (baseline)	Twitter-RoBERTa (improved)
Accuracy	0.9502 [0.9471, 0.9532]	0.9527 [0.9497, 0.9556]
Precision	0.6930	0.7217
Recall	0.6752	0.6652
F1 (toxic class)	0.6840 [0.6649, 0.7022]	0.6923 [0.6728, 0.7105]
ROC-AUC	0.9617 [0.9579, 0.9653]	0.9650 [0.9615, 0.9684]

The first thing we noticed is that the two models make different trade-offs. RoBERTa is more cautious: it wrongly flagged 410 harmless comments, compared with 478 for BERT, which raises its precision by almost 3 points. But it also missed 535 toxic comments, compared with 519 for BERT, so its recall dropped by 1 point. F1 balances precision and recall, so these two changes largely cancel out, which explains why the F1 gap is so small.



The ROC curves tell a similar story. The two lines lie almost on top of each other, with RoBERTa slightly higher at low false-positive rates, the region that matters most for moderation.



One reassuring detail: the Wilson intervals for accuracy, [0.9470, 0.9531] for BERT and [0.9497, 0.9556] for RoBERTa, are almost identical to the bootstrap intervals. Two completely different methods, one based on the binomial formula and one on resampling, give the same answer.

3.2 The difference between the models

Checking whether the two models' confidence intervals overlap is not the right test here, because both models were scored on the same comments. The paired analysis is what counts:

Metric	Difference (Δ)	95% CI	Standard error	z	p-value	Share of bootstrap samples where RoBERTa wins
F1	+0.0083	[−0.0043, +0.0210]	0.0064	1.30	0.195	90.2%
Accuracy	+0.0026	[+0.0006, +0.0046]	0.0010	2.51	0.015	99.2%
ROC-AUC	+0.0032	[+0.0011, +0.0055]	0.0011	2.88	0.004	99.8%

The F1 interval runs from −0.0043 to +0.0210. It includes zero, so a true difference of nothing at all is perfectly consistent with the data. The accuracy and ROC-AUC intervals, on the other hand, sit entirely above zero.

Why does F1 behave differently? Its standard error (0.0064) is six times larger than that of accuracy (0.0010). Accuracy is computed over all 20,000 comments, but F1 depends only on the roughly 1,600 toxic comments and the comments flagged as toxic. Fewer data points mean more noise, so F1 needs a much bigger effect, or a much bigger test set, before a difference becomes significant.

Looking at the 10,000 bootstrap differences makes this concrete. Most of them lie above zero, but about 10% fall below it. Their distribution is also almost perfectly bell-shaped (skewness −0.02), just as the central limit theorem predicts.

3.3 Hypothesis test results

Test	What it compares	Result	p-value	Decision at $\alpha = 0.05$
Permutation test (main)	F1	$\Delta F1 = +0.0083$	0.192	Fail to reject H_0
z-test with bootstrap SE	F1	$z = 1.30$	0.194	Fail to reject H_0
McNemar's test	Error rates	$b = 193$, $c = 245$, $\chi^2 = 5.94$	0.015	Reject H_0
z-test with bootstrap SE	ROC-AUC	$z = 2.88$	0.004	Reject H_0

All three methods for F1 land on $p \approx 0.19$, which gives us confidence that the result is not a quirk of one method. The permutation test shows why. The observed $\Delta F1$ of 0.0083 sits to the right of centre, but differences of that size turn up about one time in five even when the two models are interchangeable.

McNemar's test tells a different story. The models disagree on only 438 of the 20,000 comments. RoBERTa is right on 245 of these, and BERT on 193. A split like that is unlikely if the models were truly equal ($p = 0.015$), so RoBERTa does make fewer mistakes overall: 52 fewer per 20,000 comments.

3.4 What a different test set would have shown

This simulation answers the research question most directly.

Test-set size	RoBERTa wins	BERT wins	Spread of $\Delta F1$ (SD)	Power of McNemar's test
100	46.2%	33.4%	0.0983	0.2%
250	50.4%	42.2%	0.0608	1.4%
500	56.0%	40.6%	0.0413	3.5%
1,000	61.3%	37.7%	0.0289	4.8%
2,500	67.2%	32.6%	0.0186	11.6%

Test-set size	RoBERTa wins	BERT wins	Spread of ΔF1 (SD)	Power of McNemar's test
5,000	75.1%	24.9%	0.0129	21.1%
10,000	83.7%	16.3%	0.0091	40.7%
20,000	90.5%	9.5%	0.0063	66.3%

The simulation first shows how much ΔF1 jumps around from one random test set to another. With only 100 comments, it swings between about −0.2 and +0.25. RoBERTa wins only 46% of these small test sets, BERT wins 33%, and the rest are ties.

As the test set grows, the spread shrinks: from 0.098 at 100 comments to 0.0063 at 20,000, a factor of 15.6. The 1/√n rule predicts √200 ≈ 14.1, and the observed values follow it closely.

At 20,000 comments, RoBERTa wins 90.5% of the time, but BERT still wins 9.5% of the time. Even at this size, McNemar's test detects RoBERTa's advantage in only 66% of test sets, below the usual 80% target.

How big would the test set need to be? Scaling the standard error by 1/√n, detecting the F1 difference with 80% power would take about 90,000 test comments, almost the entire official test split. Detecting the accuracy difference would take about 26,000.

This was the most eye-opening part of the project for us. With a typical test set of 1,000 comments, the comparison would have been close to a coin toss, and whichever model "won" would have been largely down to chance.

3.5 Decision

On the main question, F1, we fail to reject H_0 at $\alpha = 0.05$ ($p = 0.19$). The 0.83-point improvement in F1 could plausibly be due to which comments happened to end up in the test sample.

On the supporting metrics, RoBERTa's advantage is real but small: ROC-AUC is higher by 0.0032 ($p = 0.004$) and accuracy by 0.26 points ($p = 0.015$). Both results still hold after a Bonferroni correction for testing three metrics ($\alpha/3 = 0.0167$).

So our answer is this: RoBERTa is probably slightly better at ranking comments and makes a few dozen fewer mistakes per 20,000, but its improvement in F1, the metric that matters most for flagging toxic posts, is not reliable at this test-set size. Claiming "RoBERTa is better" from the raw F1 score alone would not be justified.

3.6 Limitations

- We trained each model only once, with seed 42. A different seed would shift the scores slightly, and this training-to-training variation, which we did not measure, could be as large as the gap we are testing.

- The bootstrap and permutation tests assume that test comments are independent. Comments from the same article or thread probably are not, which would make our intervals slightly too narrow.
- The decision thresholds were chosen on 15,000 validation comments. That choice carries its own uncertainty, which our test-set intervals do not include.
- Civil Comments come from news websites, not Twitter. This mismatch may explain why the tweet-trained model helped so little; on real tweets, the gap could be larger.
- The labels are averages of human opinions cut at 0.5, so comments near that boundary have noisy labels.
- To fit within the free GPU time, we trained on 100,000 comments for 2 epochs. Both models were affected equally, but more data would probably raise both scores.
- The test had 66% power for the accuracy difference and much less for F1. A "not significant" result is therefore not proof that the models are equal; it only means this test set was not large enough to tell.

4. Where Probability and Statistics Are Used

Before this project, we thought of statistics and machine learning as two separate subjects. It turned out that nearly every step of judging a model is a statistics problem, from choosing which comments to test on to deciding whether a result can be trusted. The table below shows where each idea came in.

Concept	In simple words	Where we used it
Random and stratified sampling	Picking items by chance, or picking from each group in its true proportion	Choosing the 15,000 validation and 20,000 test comments, keeping the real 8% toxic share in the test set
Probability	A number between 0 and 1 saying how likely something is	Each model outputs the probability that a comment is toxic; above the threshold (0.84 or 0.86), it is flagged
Conditional probability	The chance of something, given that something else is true	Precision is the chance a flagged comment is really toxic; recall is the chance a toxic comment gets flagged
Binomial distribution	Counting successes in many yes/no trials	Each test comment is classified right or wrong, so accuracy is a binomial proportion
Confidence interval	A range that likely contains the true value	Every score got a 95% interval using the bootstrap, e.g. F1 difference of [−0.0043, +0.0210]
Hypothesis testing and p-value	Assume no difference, then check how surprising the data would be	H_0: both models are equally good; $p = 0.19$ for F1, so H_0 is not rejected at $\alpha = 0.05$
McNemar's test	Compares two models using only the cases where they	RoBERTa alone was right on 245 comments, BERT alone on 193; $p = 0.015$

Concept	In simple words	Where we used it
	disagree	
Sample size and standard error	Bigger samples give more stable results	The simulation showed results become more reliable as the test set grows; about 90,000 comments would be needed to detect the F1 difference

5. Conclusion

Both models turned out to be capable toxic-comment classifiers. Working on them side by side also showed us how differently they behave: RoBERTa is more cautious and raises fewer false alarms, while BERT catches slightly more toxic comments.

If we had only looked at the raw scores, we would have written "Twitter-RoBERTa beats BERT" and moved on, since it scored higher on four of the five metrics. The statistics tell a more careful story. RoBERTa ranks comments slightly better and makes 52 fewer mistakes per 20,000 comments, and both results are significant. But its F1 gain of 0.83 points could easily be due to which comments landed in the test set ($p = 0.19$).

The biggest lesson for us is that a model comparison is only as trustworthy as the test set behind it. For a difference this small, we would have needed about 90,000 test comments to detect it reliably. Reporting confidence intervals and p-values, rather than single scores, is what stops small, noisy gains from being mistaken for real progress.

If we continued this work, we would train each model several times with different random seeds to measure training variation, evaluate on the full 97,320-comment test split for more power, and test both models on real tweets, where pre-training on Twitter should matter more.